

Is the documented temperature compensation good for the prediction system?

Study 1 — the current instrument era, station 02
Medieval walls of Gubbio, static structural health monitoring

Notebook executed 11 August 2026 · Data window 21 February 2025 – 10 August 2026
· Hourly analysis grid

Summary

The preprocessing stage of the pipeline applies a fixed thermal correction to the inclinometer signal before any other stage sees it:

$$I_{\text{comp}}(t) = I(t) - (T_{\text{air}}(t) - T_{\text{air}}(t_0)) \cdot c \cdot 1000, \quad c = 0.005,$$

(1)

that is, five millidegrees of inclination removed per degree Celsius of air temperature, applied instantaneously, driven by the station’s own air-temperature channel. This report asks two questions. Does that correction help or hurt a prediction system built on this signal? And is its central premise — that the structure responds to temperature at the instant it is measured — true?

Verdict on the correction: it fails all three criteria

Criterion	Measured	Result
Thermal dependence is reduced	$ r $: 0.845 → 0.956	fail
Coefficient is near the variance-minimising value	5.00 vs 1.60 mdeg/°C	fail
Forecast improves under a controlled contrast	penalty +68.2 % to +119.3 %	fail

The measured thermal slope of this instrument is 1.63 mdeg/°C (95 % confidence interval 1.49–1.77). The documented coefficient is roughly three times that value, which places it on the far side of the variance minimum: applying it inflates the residual variance to 3.3× that of the untouched signal and 11.7× the achievable minimum, and it leaves the corrected series *more* strongly related to temperature than the raw series was — with the sign reversed.

Verdict on the premise: instantaneous in air temperature, lagged in radiation

The response to **solar radiation** is measurably delayed. Filtering the radiometer through a first-order thermal lag lifts the variance it explains from 0.575 to 0.731 at a time constant of two hours, and from 0.346 to 0.460 at four hours on the unfiltered signal. That is the physical mechanism working as expected: flux arrives, a thermal mass integrates it, the structure moves afterwards.

The response to **air temperature** shows no delay at any timescale this window can resolve. The best time constant is zero, and at the diurnal scale instantaneous air temperature explains 93.2 % of the variance of the inclination. This is not evidence against the mechanism — it is what the mechanism predicts, because air temperature is itself an already-integrated response to the same radiation and so arrives pre-lagged. Air temperature works as a proxy for the forcing precisely because it is a lagged one.

Neither finding rescues the correction, which fails on magnitude, not on timing. Window A covers half an annual cycle; the seasonal response is a separate question, and the companion legacy study answers it very differently — there the inclination *leads* air temperature by about two months, which no thermal lag can reproduce.

Recommendation. Remove the fixed compensation from Notebook 00 and let the grey-box decomposition estimate the thermal response. If a compensated series is required for presentation, the coefficient must be fitted per station, never fixed.

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1 The question and how it is decided

1.1 What is under test

Equation (1) is the correction as implemented in `heritageshm/preprocessing.py` and applied in Notebook 00. Three of its properties matter for what follows.

The coefficient is fixed. $c = 0.005$ is a constant in the source, not a fitted quantity, and it is applied identically to every station.

The response is assumed instantaneous. The correction uses the air temperature recorded at the same timestamp as the inclination. It contains no lag term. This is an assumption about the structure, and Section 4 tests it rather than accepting it.

The reference point is arbitrary. $T_{\text{air}}(t_0)$ is the first retained record, and the result is normalised so that the first value is zero. Both are constants and both vanish under first differencing, so neither affects any conclusion below.

1.2 The physical hypothesis

The wall stands in an embankment condition: one face is exposed, the other is restrained by the fill behind it. Solar radiation heats the exposed face, the heat diffuses into a large masonry mass, and the differential expansion between the two faces tilts the structure. Two consequences follow, and both are testable.

First, **the true forcing is radiation, not air temperature.** Air temperature is a proxy — the only one available wherever the radiometer is not, which is the entire legacy era and every station but this one.

Second, **a body with thermal mass cannot respond instantaneously.** Whatever the correction assumes, the structure must integrate its forcing over some characteristic time. That does not by itself mean a correction driven by air temperature must carry a lag term, because a proxy can already contain the delay. It means the delay has to be measured in whichever variable is actually used.

1.3 The three criteria for the correction

A correction that claims to remove a thermal component makes falsifiable promises. The study evaluates three, each decided by a number computed in the notebook rather than by judgement.

1. **Thermal dependence must fall.** After correction, the absolute correlation between the signal and air temperature must be smaller than before. This is the correction's own stated goal and the weakest test it could be asked to pass.
2. **The coefficient must be near the variance-minimising value.** Sweeping c traces residual variance as a function of the correction applied. A defensible coefficient sits near the minimum of that curve; one past the minimum makes the series worse than leaving it alone.
3. **The forecast must improve under a controlled contrast.** Two models of identical form, differing only in whether the target was compensated, must not be harder to forecast in the compensated case. This is the criterion that matters for the pipeline, because the pipeline exists to forecast.

1.4 Prior evidence from the manufacturer documentation

A separate review of `docs/docs-manufacturer/` established the context in which these results should be read. The instrument is a SEIKA SB1U capacitive inclinometer with an integrated conditioner: zero point 2.5 V, sensitivity approximately 1000 mV per degree, range $\pm 2^\circ$. The

logged I channel is the conditioner output in millivolts, so one millivolt corresponds to one millidegree once the 2500 offset is subtracted.

Two statements bear directly on Equation (1). First, the datasheet lists temperature-drift compensation as a *built-in* feature of the conditioner, so a software correction risks being applied on top of a hardware one. Second, the residual drift specification — 20 ppm per kelvin on the reference voltage — bounds the instrument’s own thermal error at the order of 0.1 mdeg/°C. The value applied in software is 5 mdeg/°C, roughly fifty times larger.

This also disposes of an alternative reading of the results below. The signal follows air temperature closely and without delay, which is the signature a residual instrument sensitivity would leave. It cannot be one: the measured slope is 1.63 mdeg/°C, some sixteen times the manufacturer’s bound. The thermal response being measured is the structure’s, not the sensor’s — which is exactly why it belongs in the model rather than in a preprocessing subtraction.

2 Data and method

2.1 Sampling: the hourly grid

Every environmental proxy the pipeline aligns against — ERA5-Land and the other reanalysis products — is published hourly. The record is therefore aggregated from its native 20-minute rate to one hour immediately after parsing, and every number in this report is computed on that grid. Analysing at 20 minutes and resampling afterwards would make the conclusions depend on a resolution the production pipeline never sees.

Each hour is the mean of its raw samples, kept only if at least two of the three survived cleaning, so an hour rebuilt from a single reading is not presented as equivalent to a complete one. Aggregation preserves the statistics that matter and moderates the extremes, as it should: mean air temperature is unchanged at 16.01 °C and mean inclination at −318.8 mdeg, while the largest one-off solar reading falls from 1387.8 to 1358.0 W/m². Of 38 593 native slots, 12 865 hourly intervals result, 6 of them rejected for holding too few raw samples.

One capability is lost and is restored explicitly: an hourly grid cannot resolve a lag shorter than an hour, so the native record is retained and scanned once, in Section 4, as a sub-hourly check.

2.2 Station, era and windows

Station 02, current instrument era — the only period in which wall temperature and solar radiation exist alongside the inclinometer, and therefore the only period in which the question can be asked with every candidate driver present.

Table 1: Channel coverage over the loaded span (12 865 hourly intervals).

Channel	Present	Available	Min	Max	Mean	Std
batt	9 919	77.1 %	3.271	4.964	4.161	0.396
tair	9 919	77.1 %	−5.17	38.32	16.009	8.924
rh	9 919	77.1 %	25.25	100.00	75.621	20.539
sr	8 884	69.1 %	0.00	1358.00	244.410	335.320
twall	4 706	36.6 %	1.25	55.56	23.154	11.617
inc	9 919	77.1 %	−1712.1	−261.6	−318.8	48.751

Wall temperature is the binding constraint: its probe returns the −55.0 failure sentinel across 15 566 native records, leaving it valid in only 36.6 % of the span. Two contiguous windows carry every channel simultaneously.

Table 2: The two windows in which the inclinometer, both temperatures and solar radiation are valid together.

Window	Period	Days	Complete hours	Role
A	2025-02-21 to 2025-07-13	143	3 415 of 3 432 (99.5 %)	descriptive analysis and main experiment
B	2026-06-18 to 2026-08-10	54	1 225 of 1 296 (94.5 %)	held-back out-of-period check

Window B sits eleven months after Window A, in a different season, on the far side of a 103-day outage. It is never used to choose anything; it exists only to show whether the ordering of the configurations survives a change of period.

2.3 Signal conditioning

The loader implements the contract in `docs/raw-data-format.md`: per-field decimal normalisation, dates taken from the record rather than the filename, field-wise merging of duplicate timestamps, every sentinel mapped to NaN, and the inclinometer converted from conditioner millivolts to millidegrees about its calibrated zero of 2500.

A gross-outlier screen then removes the instrument-settling transients that follow a long outage, using a rolling-median context test with the threshold set from the robust spread of the signal itself: $\sigma = 14.70$ mdeg, threshold 117.6 mdeg, 59 of 9 860 hourly samples removed (0.60 %).

2.4 The six configurations

Table 3: The configurations compared under a common model. C against D is the controlled contrast; E and F test the physics rather than the correction.

	Target	Regressors	Question it answers
A	raw	instantaneous	does the model benefit from the drivers directly?
B	compensated	instantaneous	what does correcting <i>and</i> supplying the drivers cost?
C	compensated	none	the controlled contrast, compensated arm
D	raw	none	the controlled contrast, raw arm
E	raw	lag-optimised	does shifting each driver to its best lag help?
F	raw	inertia-filtered	does a first-order thermal lag help?

Configurations E and F are built from quantities measured in Section 4, not chosen: E shifts each driver by the lag at which its levels cross-correlation peaks, and F filters each driver through the time constant that won the band-limited inertia scan. Both act on the raw target, so they compete with A and D rather than with the correction.

Every configuration is scored on the raw signal scale. Predictions of a compensated target have the correction added back before the error is computed, so all six predict the same physical quantity. Without that step the comparison would be meaningless, since a smaller mean absolute error could be bought by predicting a flatter series. The notebook asserts that the compensation is exactly invertible before any score is computed.

The controlled contrast is C against D. Window A runs from late winter to midsummer, so its test period sits largely outside the training temperature range and any configuration that

leans on temperature must extrapolate. Comparisons *between* regressor sets are therefore partly a measure of that extrapolation. C and D share the same model form and the same absence of regressors, differing only in whether the target was compensated. The verdict rests on them.

The reference model is ridge regression on 12 autoregressive lags (12 hours of history) plus the regressors, standardised, penalty $\alpha = 1$, fitted as a direct forecast at the stated horizon, with a chronological 70/30 split. Section 8.5 repeats the comparison under NeuralProphet, the model the project deploys, with a fixed random seed.

3 What the signal and its drivers look like

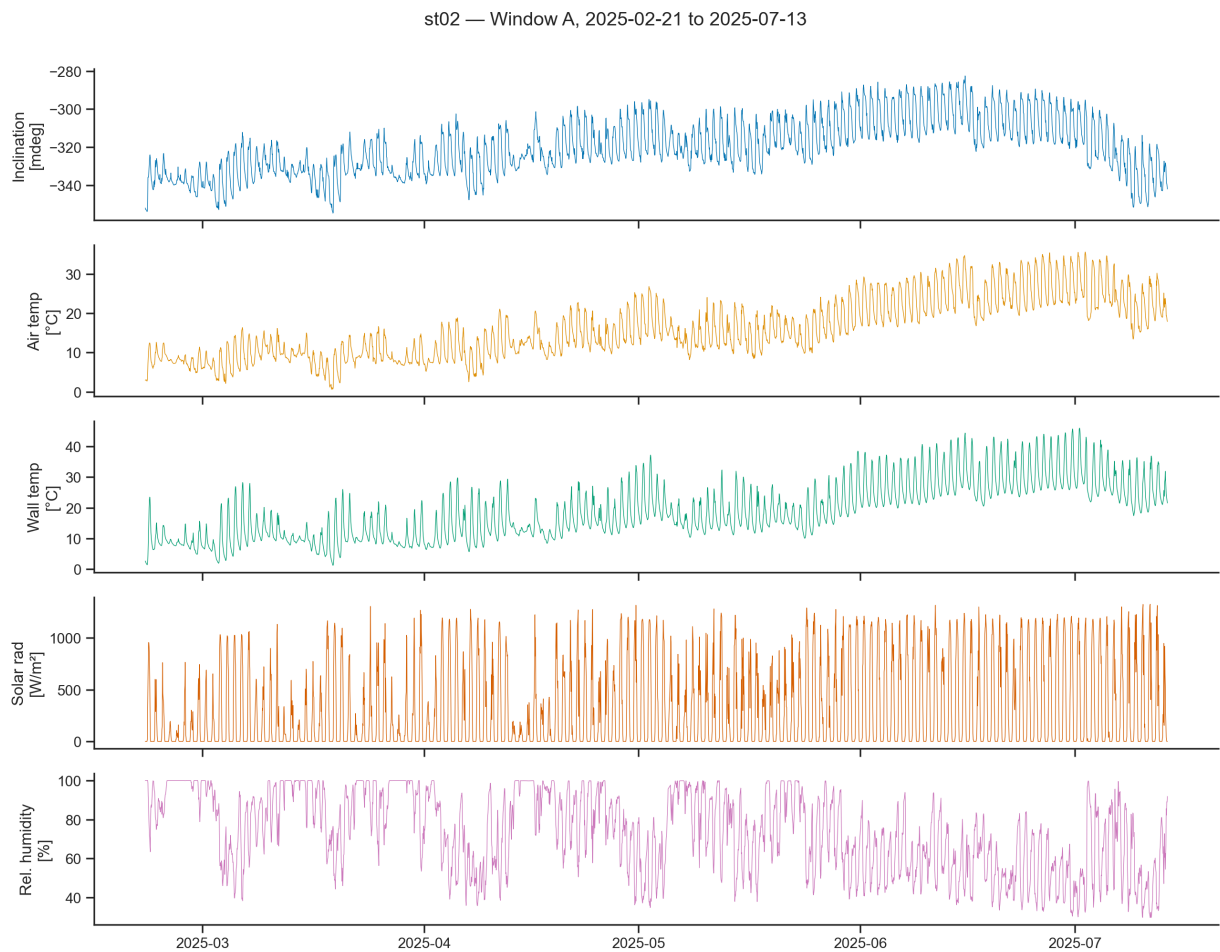


Figure 1: Window A — inclination and every candidate driver on a common time axis, 21 February to 13 July 2025.

The inclination carries a strong diurnal oscillation on top of a slow seasonal rise that follows the advance of spring into summer. Nothing in the record suggests a structural event: the series is environmental almost everywhere.



Figure 2: Window B — the held-back period, 18 June to 10 August 2026, on the far side of a 103-day outage.

3.1 Diurnal structure

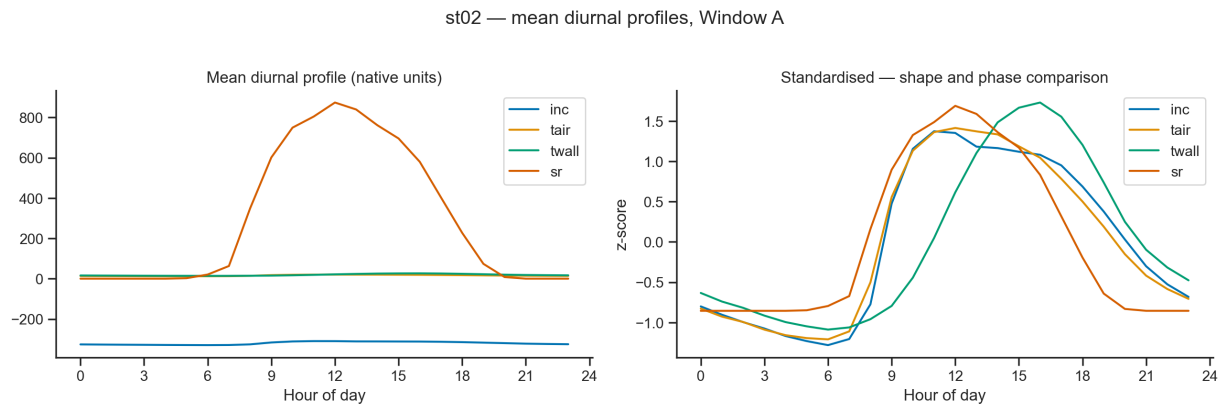


Figure 3: Mean diurnal profiles over Window A. Left, native units; right, standardised, which is the informative panel because it compares shape and phase rather than amplitude.

The standardised panel reads the phase relationships directly. The inclination and air-temperature curves rise together from about 07:00, peak together near midday and decay together through the afternoon: to the resolution of this plot they are in phase. Wall temperature behaves differently, peaking near 16:00 — the masonry probe lags the air, as it must. Solar radiation leads slightly, peaking around 12:00–13:00 with the steep morning rise characteristic of a clear-sky mean.

That the tilt tracks the *air* rather than the *wall probe* is worth holding on to: it says the element governing the tilt responds faster than the mass the probe is buried in. Section 4 makes this quantitative.

3.2 Correlation structure

Table 4: Pearson correlation of the inclination with each driver over Window A, on the hourly grid.

Driver	Levels	First differences
tair	+0.845	+0.826
twall	+0.791	+0.394
sr	+0.588	+0.398
rh	−0.611	−0.673

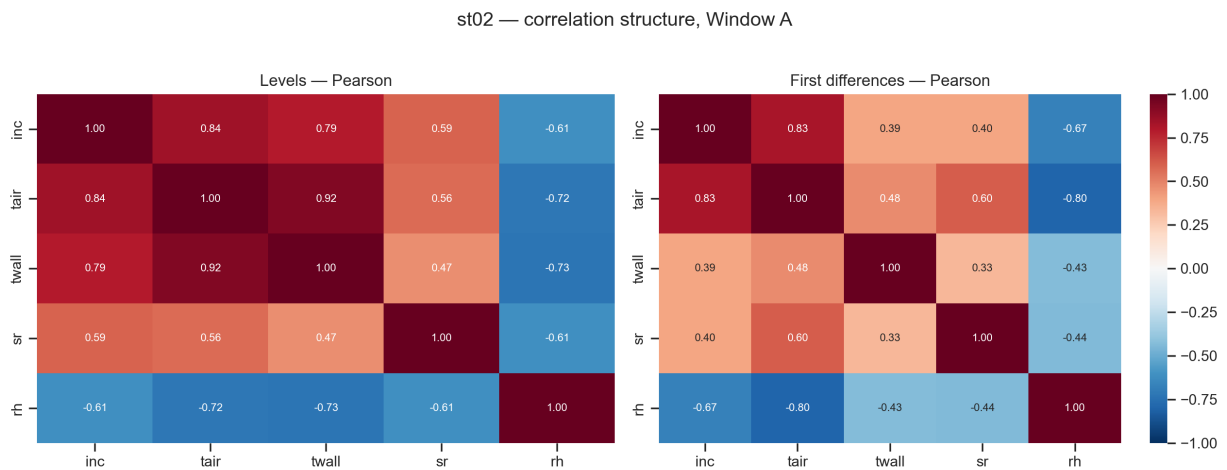


Figure 4: Correlation structure over Window A.

Air temperature is the strongest driver on both views, which justifies the correction's choice of channel even as the rest of this report rejects its magnitude. The drivers are also strongly correlated with each other — **tair** with **twall** at 0.924 in levels — so they cannot be treated as independent explanations.

The differenced correlation with **tair** is 0.826 here, against 0.375 when the same quantity is computed at 20-minute sampling. That is not a contradiction but a property of differencing: at 20 minutes the difference of a smooth signal is dominated by measurement noise, while an hourly difference is a cleaner estimate of the same physical rate of change. It is also a warning about how strongly a differenced statistic depends on the interval chosen — and Section 4 takes that warning seriously.

4 Is the response instantaneous?

A correction to an earlier version of this report

An earlier version concluded, from a single cross-correlation of *first differences at 20-minute sampling*, that “the instantaneous form of the correction is defensible”. That inference does not hold. Differencing at the sampling interval is a high-pass filter: it removes the slow component of both series by construction, so a lag-zero peak in the differenced cross-correlation shows only that the fast wiggles are synchronous. It cannot see a response governed by thermal mass, which is exactly the response in question. This section replaces that single test with four: cross-correlation on levels as well as differences, a sub-hourly check, and a scan over thermal time constants on both the full signal and the diurnal band.

4.1 Lag structure on levels and on differences

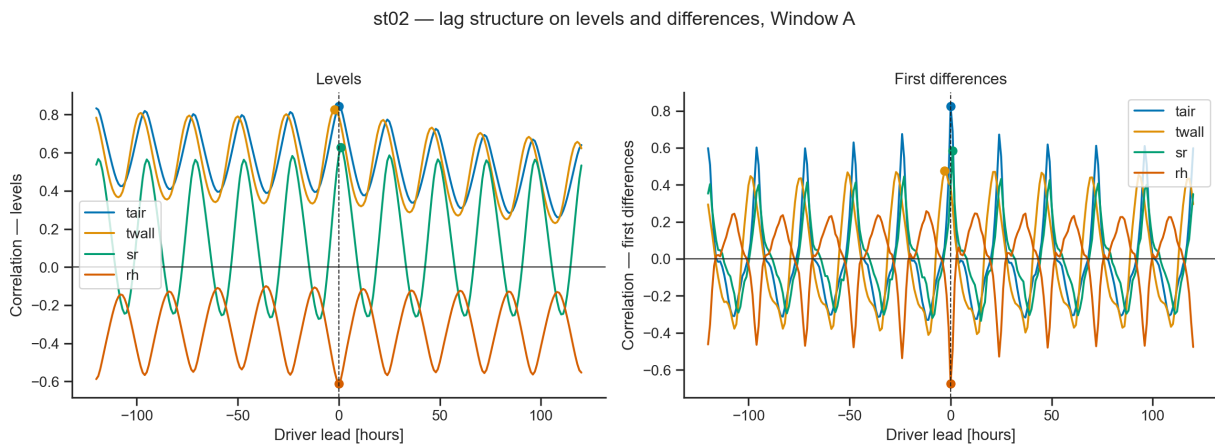


Figure 5: Cross-correlation of the inclination against each driver over ± 120 hours, on levels (left) and on first differences (right).

Table 5: Peak of each cross-correlation function. The levels search is restricted to lags at which the driver leads and whose correlation carries the sign of the contemporaneous one; without that restriction the near-periodic cross-correlation of two cyclic series returns its anti-phase extremum, which is a property of periodicity and not a lag.

Driver	Levels		First differences	
	Peak r	Lag	Peak r	Lag
tair	+0.845	0 h	+0.826	0 h
twall	+0.791	0 h	+0.477	−3 h
sr	+0.629	+1 h	+0.584	+1 h
rh	−0.611	0 h	−0.673	0 h

Air temperature peaks at zero lag on both views. Solar radiation peaks one hour ahead of the response on both. Wall temperature peaks at zero on levels but at -3 hours on differences — the inclination moves *before* the wall probe does, consistent with the probe sitting in a slower thermal mass than the one governing the tilt.

The sub-hourly check on the native 20-minute record agrees and refines the radiation figure: the differenced peak for **sr** sits at $+20$ minutes, for **tair** at zero and for **twall** at -3 hours. Nothing important hides below the analysis resolution.

4.2 Thermal inertia

A shift is a crude model of a delayed response: it assumes the structure reproduces the driver exactly, only later. A wall integrates instead. The standard lumped-capacitance model of that behaviour is a single-pole low-pass filter with time constant τ ,

$$T_{\text{eff}}(t) = T_{\text{eff}}(t - \Delta t) + \frac{\Delta t}{\tau + \Delta t} (T_{\text{driver}}(t) - T_{\text{eff}}(t - \Delta t)), \quad (2)$$

whose step response is the familiar $1 - e^{-t/\tau}$. Scanning τ and refitting at each value asks the physical question directly, and $\tau = 0$ is the instantaneous assumption — so the correction's premise sits inside the scan as a special case rather than being argued about separately.

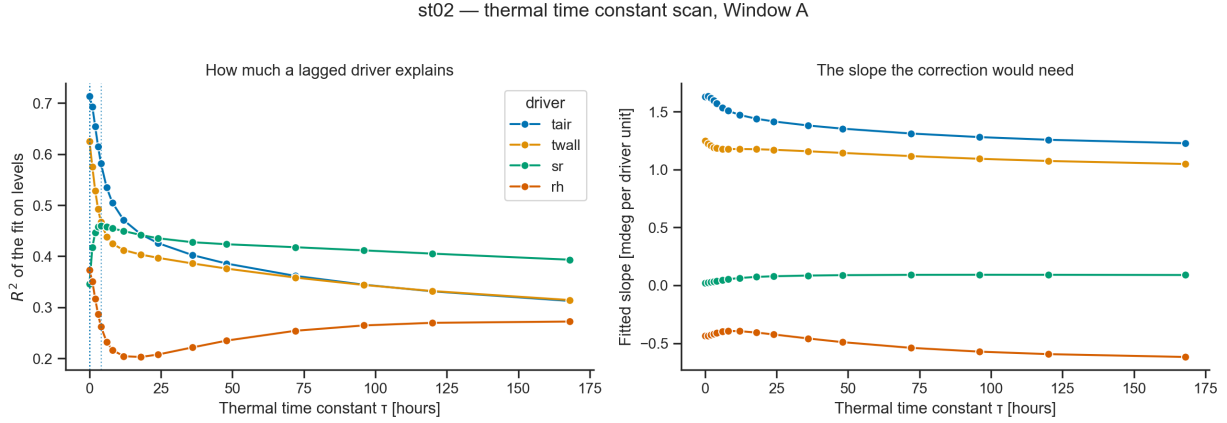


Figure 6: Explained variance and fitted slope against thermal time constant, full signal, Window A.

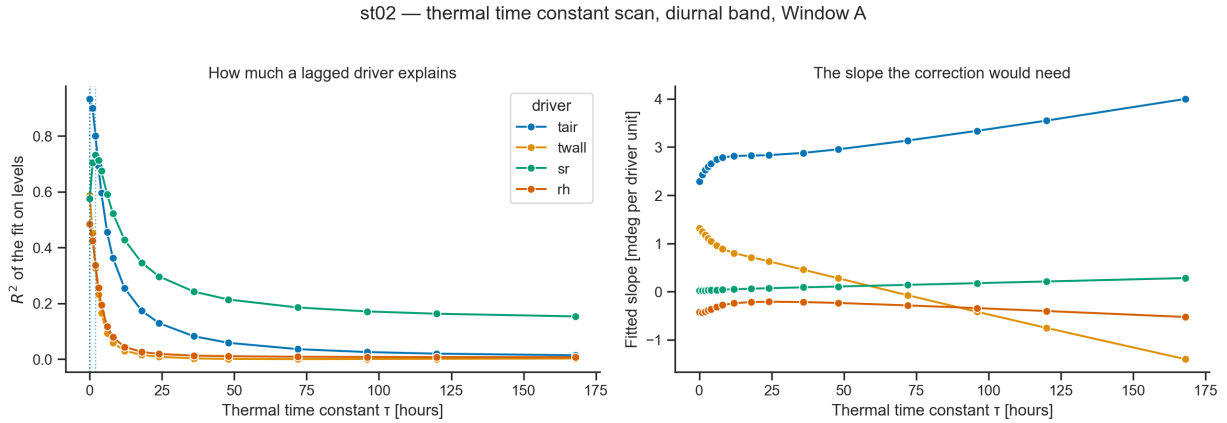


Figure 7: The same scan after removing a centred weekly mean from both series, which restricts the comparison to the diurnal band.

Table 6: Best time constant per driver. The band-limited scan removes a centred weekly mean and so describes the daily response alone; the full-signal scan is dominated by Window A's seasonal warming.

Driver	Full signal			Diurnal band		
	Best τ	R^2 , $\tau=0$	R^2 , best	Best τ	R^2 , $\tau=0$	R^2 , best
tair	0 h	0.713	0.713	0 h	0.932	0.932
twall	0 h	0.625	0.625	0 h	0.588	0.588
sr	4 h	0.346	0.460	2 h	0.575	0.731
rh	0 h	0.373	0.373	0 h	0.484	0.484

4.3 What the four tests say

Radiation acts with a lag, and the lag is physical. Filtering the radiometer through a two-hour thermal lag lifts the explained variance of the diurnal band from 0.575 to 0.731; on the full signal a four-hour constant lifts it from 0.346 to 0.460. Both are large gains bought with one parameter, and both point the same way. The mechanism of Section 1.2 is doing what it should: flux arrives, a mass integrates it, the structure moves afterwards. The one-hour peak in the cross-correlation is the same fact seen through a cruder instrument.

Air temperature acts without a lag. Every scan returns $\tau = 0$ for `tair`, and at the diurnal scale instantaneous air temperature explains 93.2 % of the variance of the inclination. Adding inertia makes the fit monotonically worse.

These two findings are consistent, not contradictory. Air temperature is itself a lagged, integrated response to the same radiation. Using it as the driver supplies the delay implicitly, which is why it needs no further filtering — and why it serves so well as a proxy in the legacy era, where no radiometer exists. What the correction gets right about timing, it gets right by accident of the proxy it happens to use.

None of this rescues the correction. Its defect is magnitude, not timing, and no choice of lag or time constant changes the results in Sections 5 to 8.

The limit of this window. Window A spans half an annual cycle, so it says nothing about the seasonal response. The companion legacy study, with a full calendar year, finds that at seasonal scale the inclination *leads* air temperature by about two months — a relationship no causal thermal lag can produce, since a filter can only delay a driver and never advance it. The daily answer given here does not generalise to the year.

5 How large is the thermal response really?

Table 7: Univariate regression of the inclination on each driver over Window A, with heteroskedasticity- and autocorrelation-consistent (Newey–West) standard errors. The comparison value is the 5.0 mdeg/°C implied by the documented coefficient.

Driver	Slope	95 % CI	SE	R^2	n
<i>Levels</i>					
<code>tair</code>	1.6279	1.4854 – 1.7705	0.0727	0.713	3 419
<code>twall</code>	1.2476	1.1443 – 1.3508	0.0527	0.625	3 419
<code>sr</code>	0.0214	0.0201 – 0.0227	0.0007	0.346	3 415
<code>rh</code>	−0.4341	−0.5111 – −0.3571	0.0393	0.373	3 419
<i>First differences</i>					
<code>tair</code>	2.1730	2.0927 – 2.2533	0.0410	0.682	3 414
<code>twall</code>	0.8394	0.7821 – 0.8967	0.0292	0.155	3 414
<code>sr</code>	0.0075	0.0066 – 0.0084	0.0005	0.158	3 407
<code>rh</code>	−0.4544	−0.4838 – −0.4249	0.0150	0.454	3 414

The documented 5.0 mdeg/°C is not inside the confidence interval and is not near it. The differenced estimate, 2.17 mdeg/°C, describes the hourly response rather than the seasonal one and is the larger of the two, but it is still less than half the documented value. No timescale available in this window produces 5.

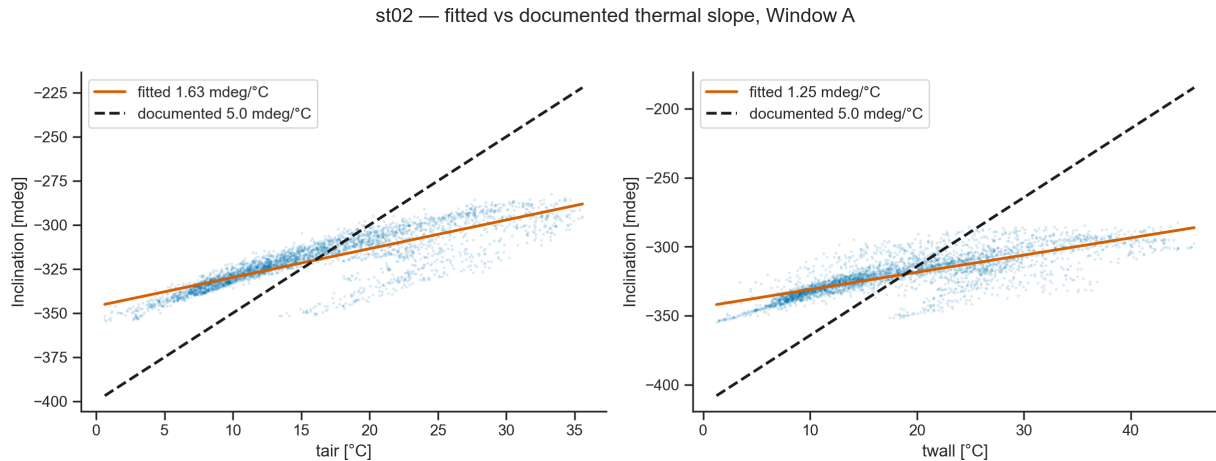


Figure 8: Inclination against air temperature (left) and wall temperature (right) over Window A, with the fitted slope and the slope implied by the documented coefficient.

The documented line is roughly three times steeper than the data. It crosses the cloud near 16 °C and departs from it in both directions, under-predicting the inclination in the cold half of the record and over-predicting it in the warm half by tens of millidegrees. The scatter also shows a second, lower branch between roughly 15 and 25 °C, displaced some 30 mdeg below the main relation: the same air temperature corresponded to a different inclination at different times, which is direct evidence that the level relation is not a single-valued function of temperature.

6 Acceptance tests

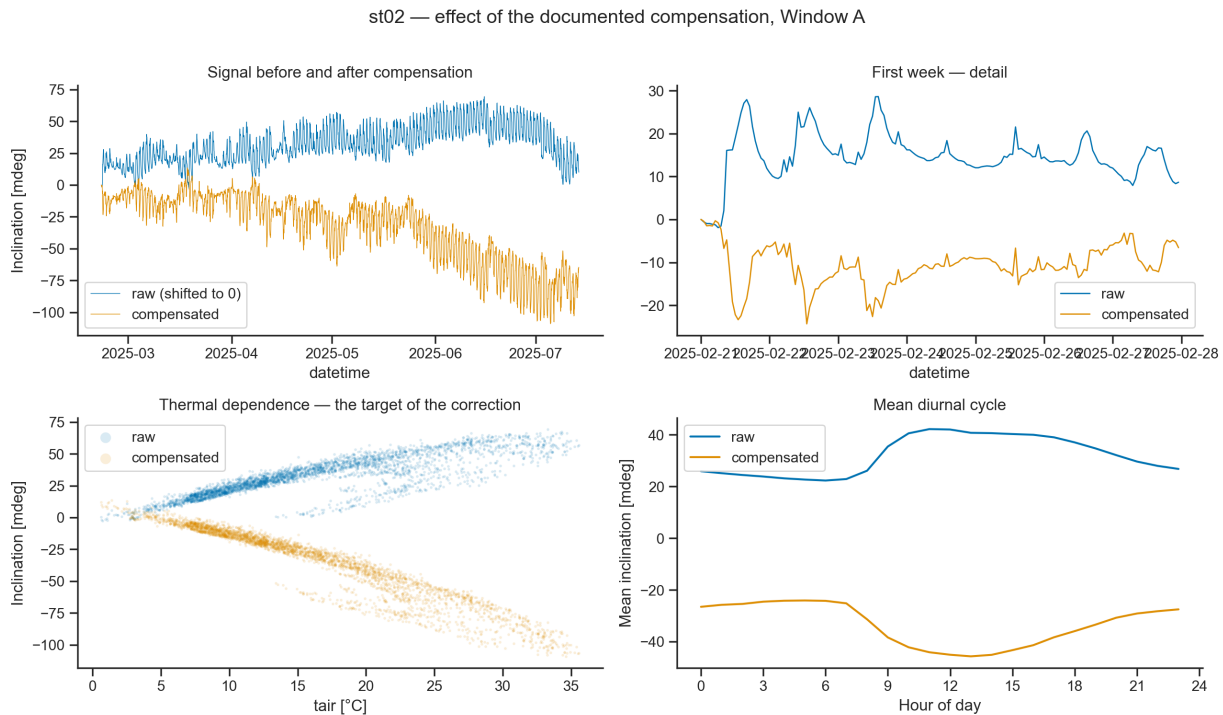
Table 8: Acceptance tests on Window A. The target for every row is a *lower* value after correction. Eight of nine fail.

Test	Raw	Compensated	Change	Verdict
Variance [mdeg ²]	211.90	709.34	+234.7 %	worse
Standard deviation [mdeg]	14.557	26.633	+83.0 %	worse
Mean diurnal amplitude [mdeg]	23.520	26.946	+14.6 %	worse
<i>r</i> with <code>tair</code> , levels	0.8446	0.9562	+13.2 %	worse
<i>r</i> with <code>tair</code> , differenced	0.8260	0.8856	+7.2 %	worse
<i>r</i> with <code>twall</code> , levels	0.7908	0.8781	+11.0 %	worse
<i>r</i> with <code>twall</code> , differenced	0.3938	0.4266	+8.3 %	worse
<i>r</i> with <code>sr</code> , levels	0.5880	0.4784	−18.6 %	better
<i>r</i> with <code>sr</code> , differenced	0.3978	0.6237	+56.8 %	worse

The single improvement is not a success. It is what happens when a series is dragged past zero: the absolute correlation with a partially independent driver passes through a minimum on its way to becoming large and negative.

Table 9: Signed correlations before and after correction. Every driver reverses sign, which is the signature of overcorrection.

Driver	Raw	Compensated	Sign reversed
tair	+0.8446	−0.9562	yes
twall	+0.7908	−0.8781	yes
sr	+0.5880	−0.4784	yes

**Figure 9:** Effect of the documented correction on Window A. Top left, the full window before and after; top right, the first week in detail; bottom left, the thermal dependence; bottom right, the mean diurnal cycle.

Each panel says something different.

Full window. The raw series rises through spring, peaks in June and turns down in July, a total excursion of roughly 60 mdeg. The compensated series does not flatten — it acquires a large downward drift, reaching about −100 mdeg by mid-July. A correction that worked would have removed the seasonal excursion; this one replaced it with a larger one of opposite sign.

First week. At weekly scale the raw and compensated traces are near mirror images about their own means.

Thermal dependence. The raw cloud slopes upward at about 1.6 mdeg/°C. The compensated cloud slopes downward at roughly −3.4 mdeg/°C — which is $1.6 - 5.0$, exactly what Equation (1) implies when the true slope is 1.6. The correction does not land near zero; it lands on the other side and further away.

Mean diurnal cycle. The raw profile swings about 20 mdeg with a midday maximum. The compensated profile swings slightly *more* and has a midday *minimum*. The daily thermal signature was not removed; it was inverted and amplified.

7 Where the documented coefficient sits

Table 10: Selected points from the coefficient sweep over Window A.

	Coefficient [mdeg/°C]	Residual variance [mdeg ²]	$ r $ with tair
No compensation	−1.0	454.66	0.931
	0.0	211.90	0.845
	1.0	83.22	0.520
	1.4	63.69	0.216
Variance-minimising	1.6	60.77	0.027
	2.0	68.62	0.339
	3.0	168.11	0.799
	4.0	381.68	0.917
Documented	5.0	709.34	0.956
	6.0	1151.08	0.973

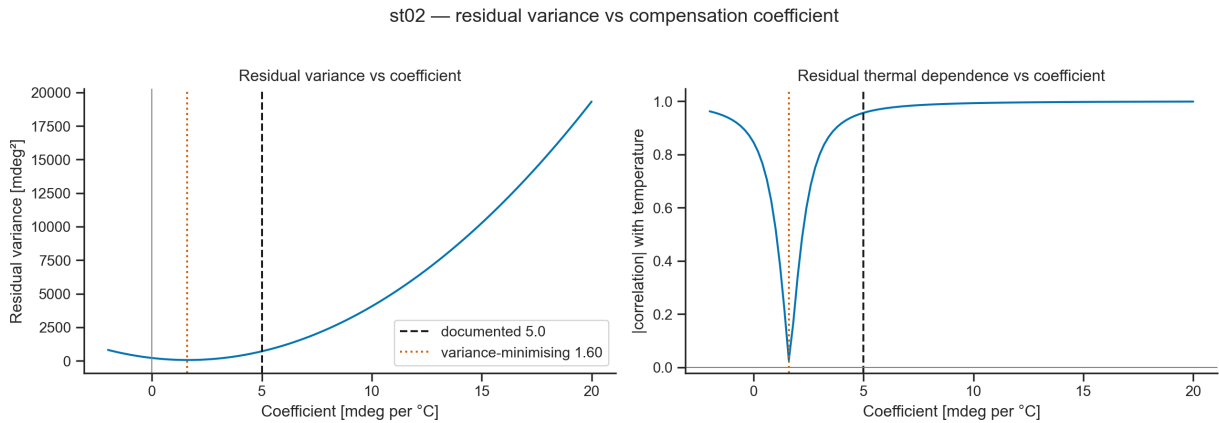


Figure 10: Residual variance and residual thermal correlation against the compensation coefficient, with the documented value marked.

Three readings decide criterion 2. First, a sharp minimum exists: at 1.6 mdeg/°C the residual variance falls to 60.8 mdeg², less than a third of the uncorrected 211.9, and the residual thermal correlation falls to 0.027. At that coefficient the correction does exactly what it claims — the mechanism is sound and only the constant is wrong.

Second, the documented value sits far past the minimum on the ascending branch: 709.3 mdeg², which is 3.3× the untouched signal and 11.7× the achievable minimum. That is the difference between a suboptimal correction and a harmful one.

Third, the variance-minimising coefficient 1.60 and the independently fitted regression slope 1.63 agree to within two per cent, from two different estimators.

8 The prediction experiment

8.1 One hour ahead

Table 11: Window A, one-hour-ahead ridge forecast, all errors on the raw signal scale. Training and test sizes are 2 351 and 1 008 samples without regressors, 2 348 and 1 007 with.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
D · raw, no regressors	1.9246	3.0875	best
C · compensated, no regressors	3.3075	5.0805	+71.9 %
A · raw + instantaneous	6.5422	8.0476	+239.9 %
F · raw + inertia-filtered	6.8054	8.4610	+253.6 %
E · raw + lag-optimised	6.8648	8.4968	+256.7 %
B · compensated + instantaneous	7.0425	8.4657	+265.9 %

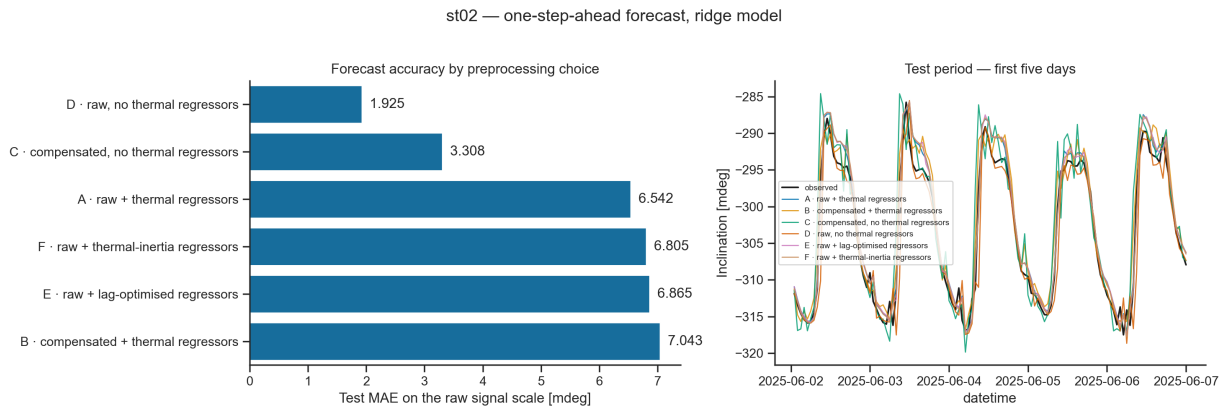


Figure 11: One-hour-ahead forecast, ridge model.

The controlled contrast is decided here: compensating the target costs 71.9 % in mean absolute error with the model form held fixed. Every configuration carrying exogenous regressors does worse still, E and F included — at a one-step horizon the previous value carries almost all the information, and Window A’s rising temperature forces any regressor-using linear model to extrapolate outside its training range. This is the confound the C-versus-D design exists to avoid.

8.2 Twenty-four hours ahead

Table 12: Window A, 24-hour-ahead ridge forecast.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
D · raw, no regressors	3.6394	5.0113	best
C · compensated, no regressors	7.6349	9.2839	+109.8 %
B · compensated + instantaneous	13.1930	16.1338	+262.5 %
F · raw + inertia-filtered	13.3373	16.1089	+266.5 %
E · raw + lag-optimised	13.5486	16.2908	+272.3 %
A · raw + instantaneous	14.1391	16.8804	+288.5 %

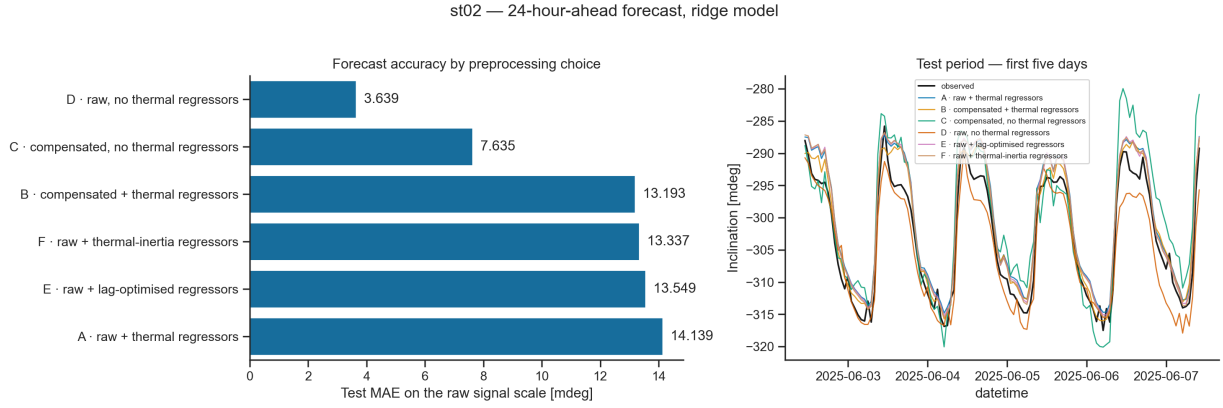


Figure 12: Twenty-four-hour-ahead forecast, ridge model.

The compensation penalty grows to +109.8%. Among the regressor-using configurations the ordering now favours the lagged and inertia-filtered forms over the instantaneous one — F and E beat A by 5.7 % and 4.2 % — the first sign that the measured lag structure carries real information, though all three remain far behind plain autoregression in this seasonally extrapolating window.

8.3 Out-of-period check on Window B

Table 13: Window B, 24-hour-ahead ridge forecast. Different season, different year, far side of a 103-day outage; 525 training and 225 test samples.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
D · raw, no regressors	3.3318	4.2758	best
E · raw + lag-optimised	5.0081	5.9162	+50.3 %
F · raw + inertia-filtered	5.2299	6.0995	+57.0 %
A · raw + instantaneous	5.7027	6.5291	+71.2 %
B · compensated + instantaneous	5.9197	7.2911	+77.7 %
C · compensated, no regressors	7.3078	9.3234	+119.3 %

In Window B the seasonal-extrapolation confound is absent — the window is midsummer through-out — and two things follow. The compensated configurations are worst, and among the regressor-using ones both lagged forms beat the instantaneous one, E by 12 % and F by 8 % relative to A. Where the model is not being asked to extrapolate, respecting the lag structure pays.

8.4 The controlled contrast, collected

Table 14: C against D across both windows and both horizons. Positive penalty means the compensated target was harder to forecast.

Window	Horizon	MAE raw (D)	MAE compensated (C)	Penalty
A	1 h	1.9246	3.3075	+71.9 %
A	24 h	3.6394	7.6349	+109.8 %
B	1 h	3.5475	5.9657	+68.2 %
B	24 h	3.3318	7.3078	+119.3 %

Four measurements, two windows, two horizons, eleven months apart. The compensated target is harder to forecast in every one of them.

8.5 Confirmation under NeuralProphet

The ridge model establishes the ordering cheaply and transparently. This repeats it under the model the project deploys — 24 autoregressive lags, 20 epochs, chronological 70/30 split, one hour ahead, fixed random seed.

Table 15: Window A, one hour ahead under NeuralProphet, 1 030 scored points per configuration, all errors on the raw signal scale.

Configuration	MAE [mdeg]	RMSE [mdeg]	vs best
A · raw + instantaneous	2.2249	2.8271	best
E · raw + lag-optimised	2.2259	2.8299	+0.0 %
D · raw, no regressors	2.2381	2.8688	+0.6 %
F · raw + inertia-filtered	2.3124	2.9605	+3.9 %
C · compensated, no regressors	3.8803	5.0989	+74.4 %
B · compensated + instantaneous	4.3028	5.6304	+93.4 %

st02 — one-step-ahead forecast, NeuralProphet

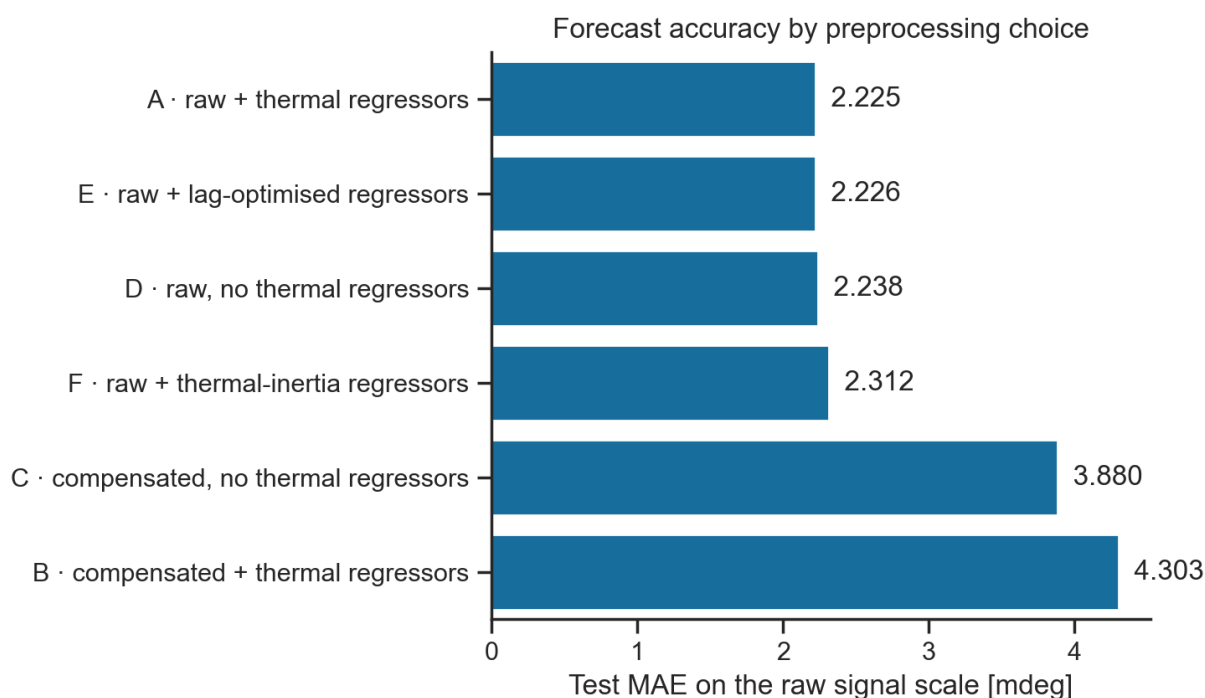


Figure 13: One-hour-ahead forecast under NeuralProphet.

The result separates cleanly into two groups. **The four configurations built on the raw signal are indistinguishable:** they span 2.22 to 2.31 mdeg, under four per cent, which is within the run-to-run variability of training this model and should be read as a tie. **The two built on the compensated signal are far worse,** by 74 % and 93 %. That gap is an order of magnitude larger than the spread among the raw configurations, and it is the finding that survives.

Training is stochastic, so the notebook fixes the seed. Seeding makes the ordering reproducible; it does not make small differences meaningful, and no claim in this report rests on one.

9 Verdict

Three criteria, three failures

- 1. Thermal dependence is not reduced.** The absolute correlation with air temperature rises from 0.845 to 0.956 and reverses sign.
- 2. The coefficient is not near the variance-minimising value.** 5.00 against 1.60 mdeg/°C, inflating residual variance to 3.3× the uncorrected signal.
- 3. The forecast does not improve.** The controlled-contrast penalty ranges from +68.2 % to +119.3 % across two windows and two horizons, and is +74.4 % under NeuralProphet. No window, horizon or model favoured the compensated target.

9.1 Recommendation

Remove the fixed compensation from Notebook 00 and let the model estimate the thermal response. Every raw configuration outperforms every compensated one under both models tested. The environmental response is exactly what the grey-box decomposition exists to

identify; subtracting a guess at it beforehand discards information the decomposition needs and injects an artefact it must then remove.

If a compensated series is wanted for presentation, the coefficient must be **fitted per station**, not fixed. At 1.60 mdeg/°C the correction does what it claims. But the legacy stations gave 7.21, 1.33 and 9.66 mdeg/°C — a sevenfold spread across three nominally identical instruments on one wall — so no single constant can serve all of them, and the constant currently in the code serves none.

9.2 On modelling the thermal response instead

The lag analysis says what should replace the correction. Air temperature can be used instantaneously, being a proxy that already carries the delay. Solar radiation should be used through a thermal lag of two to four hours rather than raw: that single parameter is worth 0.16 in explained variance at the diurnal scale. Wall temperature, as recorded by this probe, lags the tilt rather than leading it, and is the least useful of the three as a predictor whatever its value as a diagnostic.

10 Caveats

- Window A spans one half of an annual cycle. Nothing here constrains the seasonal response, and the companion legacy study shows the seasonal picture is qualitatively different from the daily one.
- The band-limited inertia scan removes a centred weekly mean, so it describes variation faster than a week. The window width is a modelling choice; a different width moves the numbers, though the ordering of the drivers is stable.
- The levels slope includes the seasonal trend and the differenced slope describes the hourly timescale. Both are reported, they differ by a factor of 1.3, and neither is near 5.0.
- The regressor-using ridge configurations are handicapped in Window A by seasonal extrapolation, which is why the verdict rests on C against D.
- Differences below about five per cent between NeuralProphet configurations are within training variability and are treated as ties.
- The manufacturer evidence in Section 1.4 comes from a separate review of the supplied documentation, not from this notebook. Every number in Sections 2 to 8 stands without it.

A Reproduction

```
conda activate neuralprophet_env
cd studies/thermal_compensation
jupyter --to ipynb thermal_compensation_study.py
jupyter nbconvert --to notebook --execute --inplace \
    thermal_compensation_study.ipynb
```

Edit the paired .py file, never the .ipynb. The first run copies 416 files out of the read-only Google Drive archive, which takes roughly forty minutes at the observed streaming rate; later runs reuse .cache/. The archive is never written to.

Setting `CONTEXT = 'paper'` in the parameter cell re-exports every figure at manuscript column size: seaborn rescales the typography and the shared canvas scale rescales the figures, so the exported figure is the same figure rather than a different one.

B Output inventory

Every figure and table in this report is regenerated by executing the notebook. The `outputs/` directory is gitignored.

Table 16: Artefacts produced by the notebook. Identifiers are stable names, not an ordering.

Identifier	Contents	In this report
S1_01	Channel coverage	Table 1
S1_02	Window A overview	Fig. 1
S1_03	Window B overview	Fig. 2
S1_04	Mean diurnal profiles	Fig. 3
S1_05, S1_06	Correlation matrices, levels and differences	Table 4
S1_07	Correlation heatmaps	Fig. 4
S1_08	Cross-correlation peaks, differences	Table 5
S1_09	Lag structure, levels and differences	Fig. 5
S1_10, S1_11	Regression slopes, levels and differences	Table 7
S1_12	Fitted versus documented thermal slope	Fig. 8
S1_13	Acceptance tests	Table 8
S1_14	Signed correlation check	Table 9
S1_15	Effect of the documented compensation	Fig. 9
S1_16	Coefficient sweep	Table 10
S1_17	Coefficient sweep figure	Fig. 10
S1_18, S1_20	Ridge results, 1 h and 24 h	Tables 11, 12
S1_19, S1_21	Experiment figures, 1 h and 24 h	Figs. 11, 12
S1_22	Ridge results, Window B	Table 13
S1_23	NeuralProphet results	Table 15
S1_24	NeuralProphet experiment figure	Fig. 13
S1_25	Verdict	Section 8
S1_26	Controlled contrast, collected	Table 14
S1_27	Cross-correlation peaks, levels	Table 5
S1_28	Cross-correlation peaks, native 20-minute record	Section 4
S1_29, S1_30	Inertia scan and best time constants, full signal	Table 6
S1_31	Inertia scan figure, full signal	Fig. 6
S1_32	Lag summary	Section 4
S1_33, S1_34	Inertia scan and best time constants, diurnal band	Table 6
S1_35	Inertia scan figure, diurnal band	Fig. 7